

Study on chemical composition, anti-inflammatory activity and quality control of the branch bark of *Morus alba* L.

Wenyuan Li^{a,b,1}, Lin Chen^{a,b,1}, Minpeng Li^{a,b}, Kaiye Peng^{a,b}, Xuemei Lin^c, Yifan Feng^{a,b}, Yun Zou^{d,**}, Xia Wu^{a,b,*}

^a New Drug Research and Development Center, Guangdong Pharmaceutical University, Guangzhou 510006, PR China

^b Guangdong Provincial Key Laboratory for Research and Evaluation of Pharmaceutical Preparations, Guangdong Pharmaceutical University, 510006 Guangzhou, PR China

^c Criminal Science and Technology Research Institute of Huizhou Public Security Bureau, Huizhou 516000, PR China

^d National Medical Products Administration Institute of Executive Development, Beijing 100073, PR China

ARTICLE INFO

Keywords:

The root bark of *Morus alba* L.
The branch bark of *Morus alba* L.
Chemical composition
Anti-inflammatory
Quality standard

ABSTRACT

In this study, the chemical constituents of branch bark and root bark of *Morus alba* L. were first qualitatively analyzed by UPLC-Q/TOF-MS. A total of 40 compounds were identified, including 20 flavonoids, 3 stilbenes, 1 phenolic acid, 1 alkaloid, 5 arylbenzofurans and 10 Diels-Alder adducts. Of these, 19 compounds were specific to the root bark, 10 to the branch bark, and 11 were common to both. Total phenolic and flavonoid contents were determined, and anti-inflammatory activities were evaluated. Ethyl acetate extracts of both significantly inhibited NO secretion, with the root bark showing a stronger effect than the branch bark. Chemical components of branch bark were further separated, and the anti-inflammatory activities of these isolated compounds were investigated. Eight compounds were isolated from the ethyl acetate extract of the branch bark of *Morus alba* L.. These compounds were able to inhibit the release of inflammatory factors, including NO, IL-6, and IL-1 β , in a concentration-dependent manner. Among them, compounds SZP-1, SZP-6 and SZP-8 exhibited superior *in vitro* anti-inflammatory activity, comparable to the positive drug meloxicam at high concentrations. In addition, the quality standard of the branch bark of *Morus alba* L. was established for the first time according to the 2020 edition of the Chinese Pharmacopoeia. This study provides valuable insights into the medicinal value of the branch bark of *Morus alba* L. and promotes the comprehensive utilization of mulberry branch resources.

1. Introduction

The mulberry tree (*Morus alba* L.) is an important medicinal plant whose root bark, twigs, leaves, and fruits are all used in medicine [1]. Mulberry root bark (Sang Bai Pi, SBP) is derived from the dried root bark of the *Morus alba* L., a species of the Moraceae family. Known for its sweet and cooling properties, it is linked with the Lung meridian and is effective in clearing the lungs, relieving asthma and promoting diuresis to reduce oedema. It is commonly used to treat conditions such as lung heat with coughing and wheezing, oedema with abdominal distension and reduced urination, and facial and skin swelling [2,3]. The mulberry branch, derived from the dried shoot of *Morus alba* L., has a bitter taste and neutral properties, and is associated with the Liver meridian. It aids

in dispersing wind-dampness and is commonly used to treat conditions such as wind-dampness paralysis, limb contractures, shoulder and arm pain, joint pain, and numbness. [3]. To date, over a hundred compounds have been isolated from mulberry plants, mainly flavonoids, stilbenes, and Diels-Alder-type adducts [4], many of which exhibit potent anti-inflammatory activity. For instance, Cudraflavone B, isolated from mulberry root bark, has been shown to inhibit TNF- α expression, block the NF- κ B signaling pathway and downregulate COX-2 expression [5]. Similarly, Sanggenon C and Sanggenon O have been shown to inhibit NO production, iNOS expression, and NF- κ B activation in LPS-stimulated RAW264.7 cells [6]. Morusin from *Morus alba* root bark has been reported to alleviate IL-1 β -induced inflammation and NF- κ B activation, providing protection against osteoarthritis [7]. Additionally,

* Corresponding authors at: New Drug Research and Development Center, Guangdong Pharmaceutical University, Guangzhou 510006, PR China.

** Corresponding author.

E-mail addresses: 52sodagreen@163.com (Y. Zou), wxiaxia@gdpu.edu.cn (X. Wu).

¹ These authors contributed equally to this work and should be considered co-first authors.

oxyresveratrol from mulberry branch bark exhibits neuroprotective effects in LPS-induced mice by alleviating neuroinflammation through the PI3K-Akt signaling pathway and modulating M1/M2 polarisation[[8].

Mulberry root bark is a common traditional Chinese medicine with various pharmacological effects [9]. In contrast, mulberry branches, the most abundant part of the tree, are often discarded as waste despite having similar secondary metabolite profiles. To date, there has been limited research on the specific phytochemical composition and pharmacological activities of mulberry branch bark, including its anti-inflammatory mechanisms.

This study conducted qualitative analyses and pharmacodynamic evaluations of the primary components in mulberry root bark and mulberry branch bark (Sang Zhi Pi, SZP), with a focus on their compositional and efficacy differences. Concurrently, the chemical constituents of mulberry branch bark were isolated, and their activities assessed. This research aims to explore the medicinal value of mulberry branch bark, supporting its broader utilization and promoting the comprehensive use of mulberry branch resources.

2. Materials and methods

2.1. Chemicals and reagents

RAW 264.7 cells were obtained from the Cell Bank of the Chinese Academy of Sciences. Fetal bovine serum (FBS), high-glucose DMEM, and penicillin-streptomycin (PSS) were purchased from Gibco (USA). Sodium nitrite was purchased from Tianjin Damao Chemical Reagent Factory. Gallic acid and aluminum nitrate were obtained from Macklin (Shanghai, China). Rutin was obtained by Sichuan Weikeqi Biological Technology Co., Ltd. Folin-Ciocalteu reagent and vanillin were purchased from Macklin. Meloxicam was purchased from Aladdin Reagent Co., Ltd. Concentrated sulfuric acid was purchased from Guangzhou Chemical Reagent Factory. Oxyresveratrol and mulberroside A were provided by Sichuan Weikeqi Biological Technology Co., Ltd. Distilled water was purchased from Watsons (Guangzhou, China). Precoated silica gel GF254 plates (Qingdao Marine Chemical Factory, Qingdao, China) were used for TLC.

2.2. Plant material

Mulberry branches were collected on April 19, 2021, in Zhenjiang, Jiangsu Province, while mulberry root bark was purchased from Lao-baixing Pharmacy, Panyu District, Guangzhou City, Guangdong Province (batch number: 201603329; origin: Jiangsu; collected on March 15, 2016). Both mulberry branches and mulberry root bark were identified by Professor Liu Jizhu of Guangdong Pharmaceutical University as belonging to the species *Morus alba* of the Moraceae family. The outer bark of the branches was peeled off and air-dried to produce mulberry branch bark. Specimens of mulberry branch bark and mulberry root bark are preserved at the Central Laboratory of Guangdong Pharmaceutical University.

2.3. General experimental procedures

UV spectra were acquired using a UV-2700 UV-Vis spectrometer (Shimadzu, Kyoto, Japan). Infrared spectra were obtained using a FT/IR-4600 spectrometer (JASCO, Japan). NMR spectra (^1H , ^{13}C HSQC, HMBC, COSY, and NOESY) were recorded on a Bruker AV-500 instrument (Bruker, USA) and subsequently processed with MestReNova 14.0 software. HR-ESI-MS data were recorded with a Waters Acquity UPLC/Q-TOF Micro MS system (Waters, USA). Mass spectra were acquired in positive and negative ion modes. The ion source parameters were set as follows: capillary voltage 3000 V; cone voltage 30 V; ion source temperature 100 °C; desolvation temperature 350 °C; desolvation gas flow rate 550 L/h; cone gas flow rate 50 L/h. The mass range was m/z 100–1000. Data acquisition and analysis were carried out using

MassLynx 4.1 software. Analytical HPLC was carried out on an ACQUITY UPLC® system (Waters, USA) with a 2487 UV detector (Waters, USA), using an ACQUITY UPLC® BEH C18 column, 2.1 × 50 mm, 1.7 µm. The column temperature was maintained at 35 °C. The mobile phase consisted of acetonitrile (B) and 0.1 % formic acid in water (A). The gradient elution conditions were set as follows: 0–5 min, 80 %–65 % A; 5–25 min, 65 %–40 % A; 25–30 min, 40 %–5 % A; 30–33 min, 5 % A; 33–35 min, 5 %–80 % A; 35–36 min, 80 % A. The injection volume was 5 µL, and the flow rate was 0.3 mL/min. Semi-preparative RP-HPLC was performed using a Waters 717 plus Autosampler (Waters, USA) and a Waters XTerra® Prep RP18 semi-preparative column (Waters, USA). Precoated silica gel GF254 plates (Qingdao Marine Chemical Factory, Qingdao, China) were used for TLC. Column chromatography (CC) was performed with silica gel (100–200 mesh and 300–400 mesh, Qingdao Marine Chemical Factory, Qingdao, China) and Sephadex LH-20 (Cytiva, USA).

2.4. Extraction and isolation

Extraction: Powdered mulberry root bark and mulberry branch bark (2 g each) were weighed accurately and placed into 100 mL round-bottom flasks. To each flask, 40 mL of 80 % ethanol was added as the extraction solvent. The samples were subjected to reflux extraction at 75 °C for 60 min. After cooling, the extracts were filtered, and the filtrates were concentrated under reduced pressure to remove ethanol. The resulting residues were dissolved in distilled water and the volume was adjusted to 50 mL in volumetric flasks. The solutions were filtered through 0.22 µm microporous membranes and analyzed by LC-MS to investigate the chemical profiles of the mulberry root bark and branch bark.

Isolation: Dried mulberry branch bark (4.5 kg) was pulverized and extracted with 95 % ethanol at a material-to-solvent ratio of 1:10 using reflux in a 70 °C water bath for 3 h, repeated three times. After cooling, the extracts were filtered through gauze and combined. The combined extract was concentrated under reduced pressure to obtain a residue (432 g), which was suspended in distilled water. The suspension was successively extracted with petroleum ether, ethyl acetate, and *n*-butanol. The solvents were recovered under reduced pressure to yield the petroleum ether (153.2 g), ethyl acetate (115.8 g), *n*-butanol (96.7 g), and water (66.3 g) fractions. A small amount of the ethyl acetate fraction was dissolved in methanol and subjected to thin-layer chromatography (TLC) to screen for suitable elution solvents. The eluent with an R_f value of 0.2 was selected for further separation. The ethyl acetate fraction was mixed with silica gel (100–200 mesh), dried, and subjected to column chromatography using silica gel (300–400 mesh) with a petroleum ether-ethyl acetate gradient (1:0 to 0:1) for elution. Fractions (1000 mL each) were collected and analyzed by TLC (UV detection, iodine staining, and vanillin-sulfuric acid staining). Fractions with similar spots were combined to yield five fractions (EA1–EA5). EA1 (3.0 g) was further separated by silica gel column chromatography using a dichloromethane-methanol gradient, yielding three sub-fractions (EA1a–EA1c). EA1b (300 mg) was purified using Sephadex LH-20 chromatography with a dichloromethane-methanol (1:1) elution, followed by semi-preparative HPLC with a 95 % methanol-water mobile phase, resulting in the isolation of SZP-1 (30 mg). EA2 (2.1 g) was subjected to silica gel column chromatography and eluted with a dichloromethane-methanol gradient, followed by semi-preparative HPLC using a 95 % methanol-water mobile phase, resulting in the isolation of and SZP-3 (12 mg). EA3 (2.4 g), EA4 (2.0 g), and EA5 (5.6 g) were each subjected to silica gel column chromatography, Sephadex LH-20 chromatography with a dichloromethane: methanol (1:1) elution, and semi-preparative HPLC using an 85 % methanol-water mobile phase. This process resulted in the isolation of SZP-2 (8 mg), SZP-5 (13 mg), SZP-4 (8 mg), SZP-6 (7 mg), SZP-7 (12 mg), and SZP-8 (6 mg), respectively.

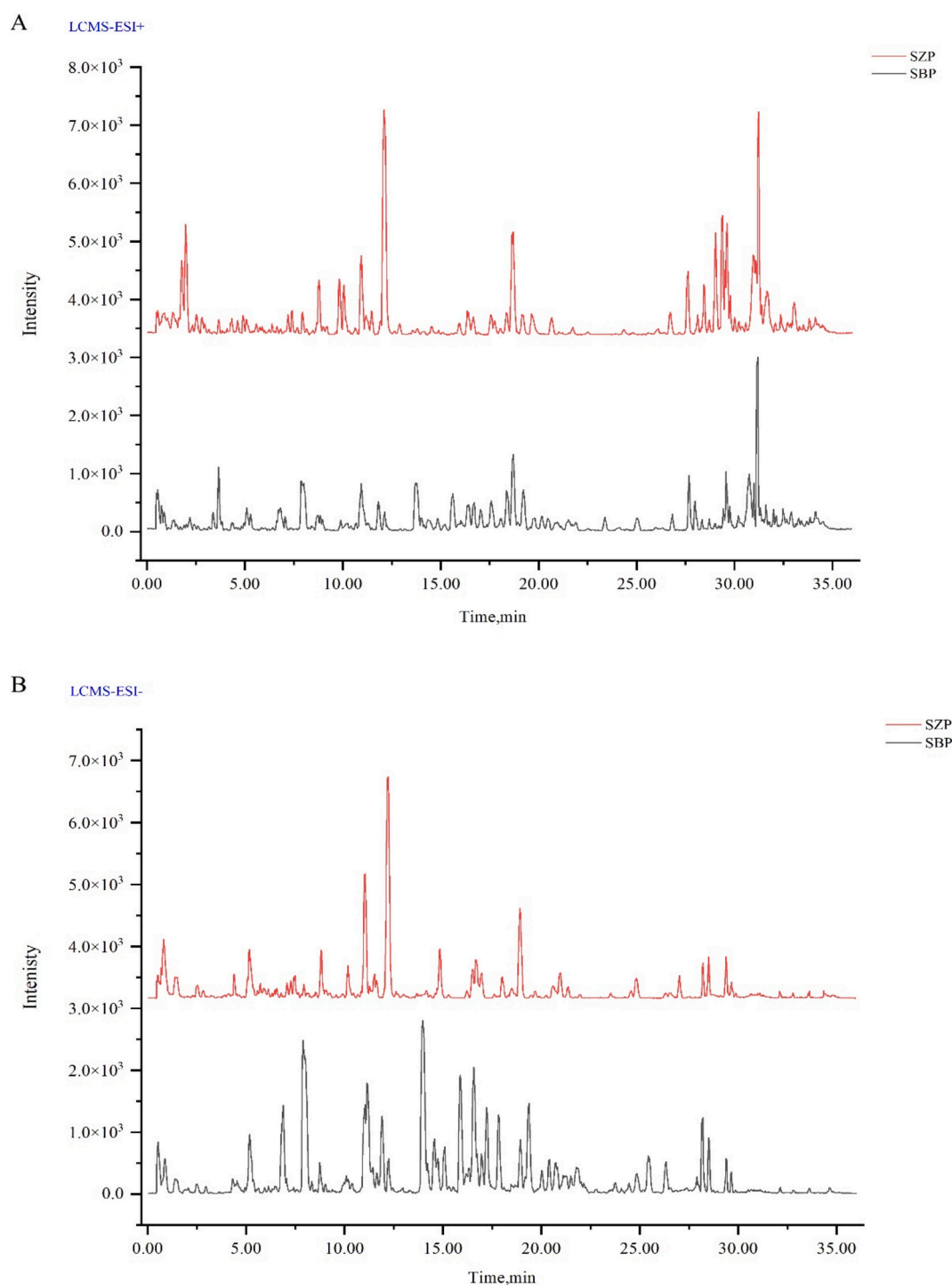


Fig. 1. Total ion chromatogram of UPLC-MS of SZP and SBP (A): Positive;(B)Negative.

2.5. Determination of total phenolic and flavonoid content

The total phenolic content was determined using the Folin-Ciocalteu method, with a gallic acid standard curve. Transfer 1 mL of the test solution into a 25 mL brown volumetric flask. Add 1 mL of Folin-Ciocalteu reagent, mix, then add 2.0 mL of 20 % Na_2CO_3 solution, and dilute to the mark with water. After standing in the dark at 25 °C for 1 h, measure the absorbance at 762 nm. Perform three parallel determinations and calculate the total phenolic content using the gallic acid standard curve.

Total flavonoid content was determined using the sodium

nitrite-aluminum nitrate method, with a standard curve of rutin. Measure 5 mL of the mulberry root bark test solution and 10 mL of the mulberry branch bark test solution into separate 25 mL brown volumetric flasks. Add 1 mL of 5 % NaNO_2 solution, mix, and let stand for 6 min. Add 1 mL of 10 % $\text{Al}(\text{NO}_3)_3$ solution, shake, and let stand for 6 min. Add 10 mL of 5 % NaOH solution, dilute to volume with water, shake, and let stand for 15 min. Measure the absorbance at 510 nm using water as the blank. Perform three parallel determinations and calculate the total flavonoid content using the rutin standard curve.

Table 1
Compounds identified from the ethanol extracts of SZP and SBP by UPLC-Q-TOF-MS.

NO.	RT (min)	Adducts	Observed (m/z)	Theoretical (m/z)	Error/ ppm	MS/MS Fragment	Molecular Formula	Identification	Category	Source
1	0.5	[M-H] ⁻	567.1692	567.1714	0.4	405.1179, 243.0593	C ₂₆ H ₃₂ O ₁₄	Mulberroside A [10]	Stilbenes	SZP
2	0.55	[M-H] ⁻	191.0549	191.0556	-0.4	171.0316, 156.7950, 127.031	C ₇ H ₁₂ O ₆	Quinic acid [11]	Phenolic acid	SBP
3	0.59	[M-H] ⁺	164.0929	164.0923	0.4	128.0697, 245.0635	C ₆ H ₁₃ NO ₄	1-Deoxynojirimycin [12]	Alkaloids	SZP, SBP
4	0.84	[M-H] ⁺	407.1224	407.1342	-2.9	245.0635, 227.06, 217.0745	C ₂₀ H ₂₂ O ₉	Oxyresveratrol-2-O-β-D-glucopyranoside [13]	Stilbenes	SZP
5	0.85	[M-H] ⁻	463.1047	463.0877	3.7	301.0547, 273.0459	C ₂₁ H ₂₀ O ₁₂	Isoquercitrin [11]	Flavonoids	SBP
6	1.41	[M-H] ⁻	243.0611	243.0657	1.9	225.0449, 201.0695, 185.0533	C ₁₄ H ₁₂ O ₄	Oxyresveratrol [14]	Stilbenes	SZP, SBP
		[M + H] ⁺	245.0736	245.0814	3.2	227.1793, 203.1358				
7	2.48	[M-H] ⁻	241.0455	241.0501	-1.9	199.0383, 273.0773	C ₁₄ H ₁₀ O ₄	Moracin M [14]	Aryl benzofuran	SZP, SBP
8	2.93	[M-H] ⁻	325.1082	325.1076	0.2	273.0773, 241.1137	C ₁₉ H ₁₈ O ₅	Moracin P [15]	Aryl benzofuran	SBP
9	4.36	[M + H] ⁺	327.1365	327.1232	4.1	275.1793, 245.0635				
		[M-H] ⁻	325.1076	325.1076	0	253.0369, 211.1319	C ₁₉ H ₁₈ O ₅	Moracin O [16]	Aryl benzofuran	SBP
10	6.76	[M-H] ⁻	711.2231	711.2442	3	601.1653, 549.1937, 439.1639	C ₄₀ H ₄₀ O ₁₂	Sanggenon T [15]	Diels-Alder-type adducts	SBP
		[M + H] ⁺	713.2462	713.2598	-1.9	695.2738, 603.2430				
11	7.46	[M-H] ⁻	579.1551	579.1655	-1.8	469.119, 451.129, 359.0868	C ₃₄ H ₂₈ O ₉	Mulberrofuran C [14]	Diels-Alder-type adducts	SZP
		[M + H] ⁺	581.1933	581.1812	2.1	471.0874, 453.1075				
12	7.86	[M-H] ⁻	625.1492	625.171	-3.5	607.1425, 515.1169	C ₃₅ H ₃₀ O ₁₁	Kuwanon L [15]	Diels-Alder-type adducts	SBP
		[M + H] ⁺	627.1835	627.1866	-0.5	609.1784, 517.1648				
13	8.83	[M-H] ⁻	437.1393	437.16	-4.7	419.1493, 367.1382, 327.1109, 245.0587	C ₂₅ H ₂₆ O ₇	Sanggenol F [17]	Flavonoids	SZP
		[M + H] ⁺	439.1932	439.1757	4	421.0932, 368.0861				
14	8.67	[M-H] ⁻	581.1668	581.1812	-2.5	563.1385, 471.1419, 361.0889	C ₃₄ H ₃₀ O ₉	Kuwanon Y [18]	Diels-Alder-type adducts	SBP
15	10.18	[M-H] ⁻	437.1496	437.16	-2.4	367.1382, 327.1109	C ₂₅ H ₂₆ O ₇	Mulberranol [14]	Flavonoids	SZP
		[M + H] ⁺	439.1792	439.1757	0.8	421.0932, 365.0739				
16	10.28	[M + H] ⁺	439.1592	439.1757	-3.8	383.0667, 365.0646, 311.0211	C ₂₆ H ₃₀ O ₆	Kuwanon U [19]	Flavonoids	SZP
		[M-H] ⁻	691.2026	691.2179	-2.2	581.1786, 471.1843, 353.0953				
17	11.04	[M + H] ⁺	693.2134	693.2336	-2.9	581.1369, 355.1088	C ₄₀ H ₃₆ O ₁₁	Kuwanon G [14]	Diels-Alder-type adducts	SZP, SBP
18	11.17	[M-H] ⁻	707.1846	707.2129	-4	689.1713, 638.1253, 353.1045	C ₄₀ H ₃₆ O ₁₂	Sanggenon C [15]	Diels-Alder-type adducts	SZP, SBP
19	11.87	[M-H] ⁻	693.2125	693.2179	-0.8	675.2136, 567.1797	C ₄₀ H ₃₈ O ₁₁	Sanggenon G [15]	Diels-Alder-type adducts	SBP
		[M + H] ⁺	695.2487	695.2492	-0.1	677.2118, 585.3035				
20	12.25	[M-H] ⁻	421.1518	421.1651	-3.2	351.0789, 311.2123, 241.0455	C ₂₅ H ₂₆ O ₆	Cudraflavone C [20]	Flavonoids	SZP, SBP
		[M + H] ⁺	423.1928	423.1808	2.8	353.1121, 311.0371, 241.0343				
21	14.01	[M-H] ⁻	693.2113	693.2336	-3.2	583.1823, 531.1833, 287.0472, 259.1287	C ₄₀ H ₃₈ O ₁₁	Kuwanon O [13]	Diels-Alder-type adducts	SBP
22	14.59	[M + H] ⁺	695.2484	695.2492	-0.1	677.2382, 585.1970	C ₄₅ H ₄₄ O ₁₁	Kuwanon H [19]	Diels-Alder-type adducts	SBP
		[M-H] ⁻	759.2469	759.2805	-4.4	581.1668, 515.1392, 353.077				
23	14.85	[M + H] ⁺	761.2881	761.2962	-1.1	581.1668, 517.2317, 355.2527	C ₂₄ H ₂₆ O ₄	Albafuran A [21]	Aryl benzofuran	SZP
		[M-H] ⁻	377.1643	377.1753	-2.9	254.0558, 186.06				
24	15.03	[M-H] ⁻	757.2685	757.2649	0.5	647.3317, 355.1176	C ₄₅ H ₄₂ O ₁₁	Kuwanon W [22]	Flavonoids	SBP
		[M + H] ⁺	759.3036	759.2805	3.0	649.1531, 357.1358				

(continued on next page)

Table 1 (continued)

NO.	RT (min)	Adducts	Observed (m/z)	Theoretical (m/z)	Error/ppm	MS/MS Fragment	Molecular Formula	Identification	Category	Source
25	15.88	[M-H] ⁻	707.1986	707.2129	-2	689.197, 638.1154, 597.173, 353.0953	C ₄₀ H ₃₆ O ₁₂	Sanggenon D [15]	Diels-Alder-type adducts	SBP
26	16.55	[M-H] ⁻ [M + H] ⁺	423.1878 425.1573	423.1808 425.1694	1.7 -2.8	405.1768, 297.139 407.1920 351.1302	C ₂₅ H ₂₈ O ₆	Sanggenol A [14]	Flavonoids	SBP
27	18.98	[M-H] ⁻	419.1367	419.1495	-3.1	365.082, 350.0637, 309.1046	C ₂₅ H ₂₄ O ₆	Cyclomulberrin [14]	Flavonoids	SZP, SBP
28	19.33	[M-H] ⁻	407.1868	407.1858	0.2	255.1704	C ₂₅ H ₂₈ O ₅	Kuwanol D [13]	Flavonoids	SBP
29	20.02	[M-H] ⁻	421.1492	421.1651	-3	403.1147, 351.0606, 311.152, 243.1127	C ₂₅ H ₂₆ O ₆	Kuwanon C [15]	Flavonoids	SZP, SBP
30	21.17	[M-H] ⁻	421.1707	421.1651	1.3	311.1264	C ₂₅ H ₂₆ O ₆	Sanggenol L [14]	Flavonoids	SBP
31	21.48	[M-H] ⁻	507.2316	507.2383	-1.3	437.1292, 367.0539	C ₃₀ H ₃₆ O ₇	Sanggenol D [23]	Flavonoids	SBP
32	22.14	[M-H] ⁻	419.1493	419.1495	0.1	350.0728, 309.1133	C ₂₅ H ₂₄ O ₆	Morusin [14]	Flavonoids	SZP, SBP
33	22.33	[2 M-H] ⁻ [2 M-H] ⁺	839.3029 841.3296	839.3068 841.3224	-0.5 0.9	419.1593, 297.2321 421.1232 301.1204	C ₅₀ H ₄₈ O ₁₂	Morusin dimer [24]	Flavonoids	SZP
34	22.81	[M-H] ⁻ [M + H] ⁺	391.1901 393.1948	391.1909 393.2066	-0.2 -3	307.1069, 253.0447 254.8978	C ₂₅ H ₂₈ O ₄	Mulberrofuran B [25]	Aryl benzofuran	SZP, SBP
35	23.76	[M-H] ⁻	489.2136	489.2277	-2.9	419.8396, 309.0274	C ₃₀ H ₃₄ O ₆	Rubraflavone C [26]	Flavonoids	SBP
36	24.54	[M-H] ⁻	417.1416	417.1338	1.9	373.1621, 262.8429	C ₂₅ H ₂₂ O ₆	Cyclomorusin [23]	Flavonoids	SZP
37	25.43	[M-H] ⁻	491.2337	491.2434	-2	473.2216, 403.3108	C ₃₀ H ₃₆ O	Sanggenol C [27]	Flavonoids	SZP, SBP
38	26.32	[M-H] ⁻	491.2273	491.2434	-3.3	473.2648, 261.1272	C ₃₀ H ₃₆ O ₆	Sanggenol P [14]	Flavonoids	SBP
39	27.01	[M-H] ⁻	303.0412	303.0505	-3.1	275.2044	C ₁₅ H ₁₂ O ₇	Dihydromorin [14]	Flavonoids	SZP
40	27.92	[M-H] ⁻	489.2339	489.2277	1.3	471.1419, 261.1351	C ₃₀ H ₃₄ O ₆	Sanggenol B [23]	Flavonoids	SBP

2.6. Quality control

The moisture, total ash, and extractives content of mulberry branch bark were determined according to the 2020 edition of the Chinese Pharmacopoeia. The moisture content was measured using the drying method, the total ash content was determined gravimetrically, and the ethanol extractives were quantified as a percentage of the sample weight. Thin-layer chromatography (TLC) was used to identify the samples. Two grams of mulberry branch bark powder were extracted with 20 mL of methanol through ultrasonic extraction (300 W, 30 min). The extraction was repeated three times, and the supernatants were combined and concentrated under reduced pressure. The residue was dissolved in 1 mL of methanol to prepare the test solution. Morusin was used as the reference standard. The TLC analysis was performed on silica gel GF254 plates with dichloromethane-methanol (19:1) as the mobile phase. The spots were observed under UV light at 254 nm. The oxyresveratrol content in mulberry branch bark was determined by HPLC. The sample solution was prepared by extracting 0.5 g of mulberry branch bark powder with 20 mL of 75 % ethanol under reflux at 75 °C for 60 min. After filtration and concentration, the residue was reconstituted in 75 % ethanol to a final volume of 10 mL. The standard solution was prepared by dissolving oxyresveratrol in ethanol to a concentration of 1 mg/mL. Chromatographic analysis was conducted using a ChromCore AQ C18 column (5 μm, 4.6 × 250 mm) with a gradient mobile phase consisting of methanol (A) and 0.2 % formic acid (B). The gradient program was as follows: 20–22 % A from 0 to 2 min, 22–35 % A from 2 to 10 min, 35–94 % A from 10 to 34 min, followed by re-equilibration to initial conditions. The flow rate was maintained at 1.0 mL/min, the column temperature at 30 °C, and the detection wavelength was set to 329 nm. The method was validated for linearity, precision, repeatability, stability, and accuracy. A calibration curve was established using five concentrations of the oxyresveratrol standard, with the linear regression coefficient confirming strong linearity. Precision was evaluated by injecting the same sample solution six times consecutively, while repeatability was assessed by preparing six independent replicates of the sample. Stability was tested by analyzing the

same solution at specified time intervals (0, 2, 4, 8, 10, and 12 h). Accuracy was determined through recovery tests, where known amounts of oxyresveratrol standard were spiked into the samples, and the recovery percentages were calculated. Finally, the oxyresveratrol content in 11 batches of mulberry branch bark samples was measured using the validated method.

2.7. Cytotoxicity assay

RAW 264.7 cells in the logarithmic growth phase were seeded in 96-well plates (2 × 10⁴ cells/well) with six replicates per concentration, including zeroing, blank control, and treatment groups. After 24 h, the medium was replaced with fresh medium containing different drug concentrations (1.2, 3, 6, 12 μg/mL), 100 μL per well. The remaining wells received complete medium. After another 24 h, 100 μL of MTT solution was added to each well and incubated in the dark for 4 h. The supernatant was discarded, 100 μL of DMSO was added, and the plates were shaken at 300 r/min for 10 min. Absorbance was measured at 570 nm to calculate cell viability.

2.8. Anti-inflammatory activity

RAW 264.7 cells were seeded in 96-well plates at 2 × 10⁴ cells/well. After 24 h, the wells were divided into blank, model, and treatment groups. One hour after drug administration, 100 μL of LPS (2 μg/mL) was added to the model and treatment groups, while the blank group received complete medium. After 24 h, supernatants were collected. Some were stored at -20 °C for IL-6 and IL-β detection using ELISA kits. The rest were used to measure NO content by transferring them to a new 96-well plate with NaNO₂ standards. Then, 50 μL of Griess reagent A and 50 μL of Griess reagent B were added. Absorbance was measured at 540 nm.

2.9. Statistical analysis

Data were expressed as mean ± standard deviation (SD). Statistical

Table 2

Total polyphenol and flavonoid content in ethanol extracts from different sections of SBP and SZP.

Extraction segmentation	Total polyphenol content (μg/mg)		Total Flavonoid content (μg/mg)		Combined total (μg/mg)	
	SBP	SZP	SBP	SZP	SBP	SZP
All	102.66	83.33	120.18	113.54	222.84	196.87
Petroleum ether extraction	56.86	83.38	159.95	110.23	216.82	193.61
Ethyl acetate extraction	321.15	180.70	365.53	216.21	686.67	396.91
n-Butanol extraction	93.26	81.92	86.80	72.65	180.06	154.56
Aqueous phase extraction	10.64	8.75	17.19	12.08	27.83	20.84

analysis was performed using GraphPad Prism 8.0 software. Differences between groups were analyzed using *t*-tests and ANOVA. Significance was considered as follows: $p < 0.001$ (extremely significant), $p < 0.01$ (very significant), $p < 0.05$ (significant), and $p > 0.05$ (not significant).

3. Result and discussion

3.1. Identification of compounds from mulberry branch bark and mulberry root bark by UPLC-Q-TOF-MS

Mulberry root bark and branch bark are derived from the root and branches of the mulberry tree, respectively. Although mulberry root bark is the primary source for clinical applications, the chemical composition of mulberry branch bark remains less explored. To compare the chemical compositions of mulberry branch bark and mulberry root bark, this study employed a UPLC-Q-TOF-MS method to analyze the ethanolic extracts and identify compositional differences. The base peak chromatogram (BPC) of ethanol extracts from SBP and SZP in negative and positive ion modes is shown in Fig. 1. In total, 40 components were identified. These include 20 flavonoids, 3 stilbenes, 1 phenolic acid, 1 alkaloid, 5 aryl benzofurans, and 10 Diels-Alder adducts. Among the identified constituents, 30 compounds were identified from SBP, of which 19 were unique to SBP. These unique compounds include various flavonoids, Diels-Alder adducts, and aryl benzofurans, such as moracin

P, kuwanon L, and rubraflavone C. In contrast, SZP contained 21 identified compounds, with 10 compounds found exclusively in SZP, such as mulberroside A, mulberranol, and kuwanon U. Detailed information on the identified compounds is provided in Table 1, and their structural formulas are presented in Supplementary Information.

3.2. Identification of anti-inflammatory active fractions in mulberry branch bark and mulberry root bark

Flavonoids are the main active compounds in the branches of *Morus nigra* L. [28]. Research has also found that flavonoids and phenolics play an important role in the anti-inflammatory effects of *Morus* species [29,30]. LC-MS analysis identified flavonoids and phenolics as the major chemical constituents of both mulberry branch bark and root bark. To further investigate their anti-inflammatory potential, we quantified the total phenolic and flavonoid contents in ethanolic extracts of different parts of mulberry root and branch bark and evaluated their anti-inflammatory activity.

3.2.1. Extraction segmentation and quantification of total phenolics and total flavonoids in the ethanol extracts of mulberry root bark and mulberry branch bark

The concentrations of total phenolics and flavonoids in the ethanolic extracts of mulberry branch and root barks, fractionated by different solvents, are ranked in descending order as follows: ethyl acetate > petroleum ether > n-butanol > aqueous phase. The results (Table 2) show that among the different solvent fractions of the ethanolic extracts of SBP and SZP, the ethyl acetate fraction contained the highest concentrations of total flavonoids and phenolics. Moreover, the total flavonoid and phenolic contents in the ethyl acetate fraction of SBP were higher than those in SZP.

3.2.2. Effect of different extraction fractions of ethanolic extracts of SBP and SZP on LPS-induced NO release in RAW264.7 cells

The Griess method was used to evaluate the effects of different extraction fractions (ethyl acetate, petroleum ether, n-butanol, and aqueous fractions) of the ethanolic extracts of SBP and SZP on LPS-induced NO production in RAW 264.7 cells, at various concentrations. The results (Fig. 2) showed that the ethyl acetate fractions of both SBP and SZP significantly inhibited NO production, with SBP exhibiting a

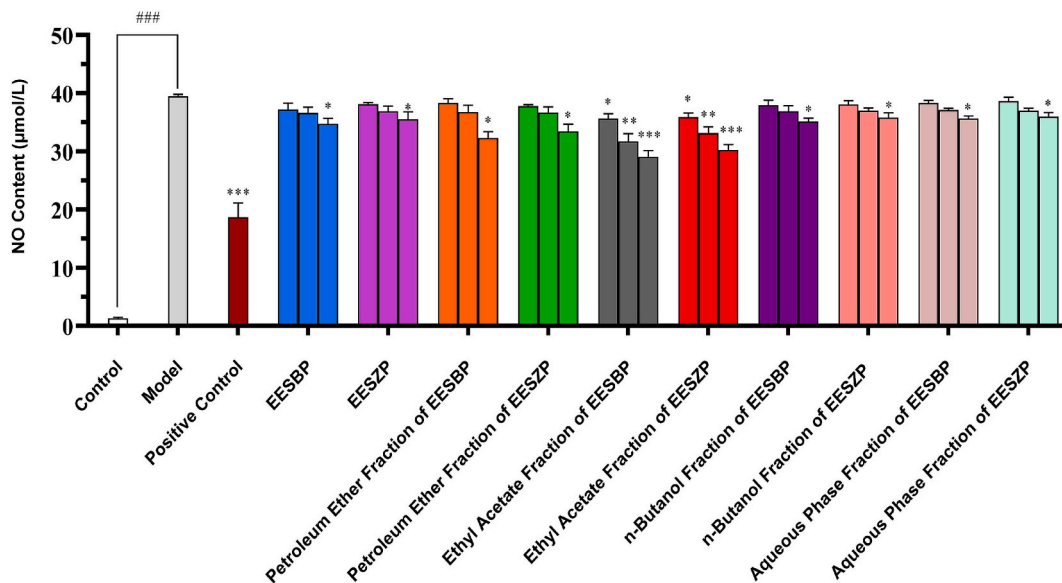


Fig. 2. Effect of extraction fractions of ethanolic extracts from SBP and SZP on LPS-induced NO content in RAW 264.7 cells. ### $p < 0.001$, vs. the control group. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, vs. the model group. EESBP: Ethanolic Extract of SBP, EESZP: Ethanolic Extract of SZP. Extract concentrations: 1.2, 6, and 12 μg/mL from left to right.

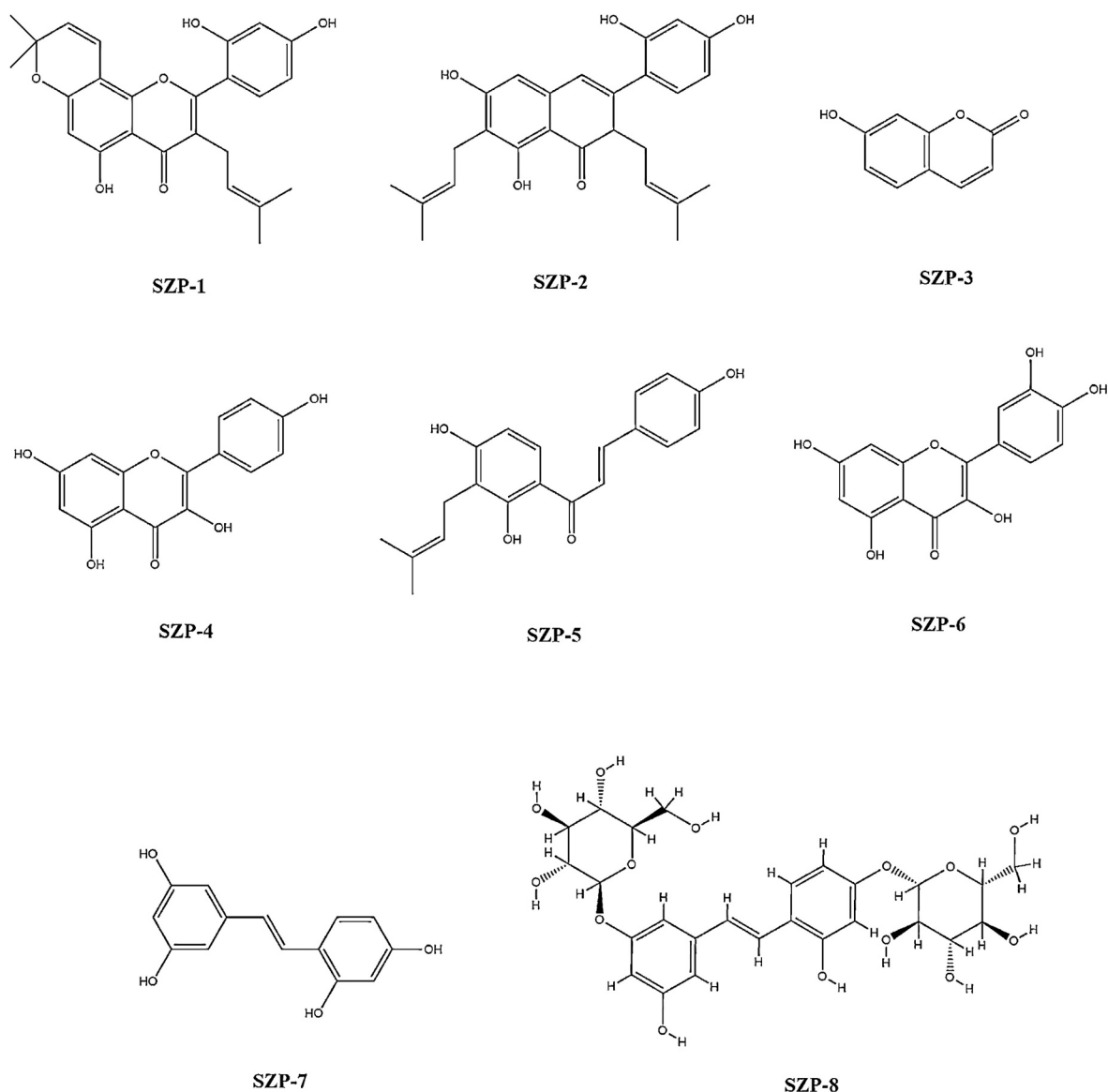


Fig. 3. Chemical composition of the ethyl acetate fraction from EESZP.

slightly stronger inhibitory effect compared to SZP. These results suggest that the ethyl acetate fraction is the major active fraction responsible for the anti-inflammatory effects of both SBP and SZP.

3.3. Isolation of anti-inflammatory active components from mulberry branch bark

To investigate the specific chemical composition of the ethyl acetate fraction of EESZP, which exhibits strong anti-inflammatory activity, we isolated and identified eight known compounds, as shown in Fig. 3. These compounds include morusin (SZP-1), cudraflavone C (SZP-2), umbelliferone (SZP-3), kaempferol (SZP-4), isobavachalcone (SZP-5), quercetin (SZP-6), oxyresveratrol (SZP-7), and mulberroside A (SZP-8). Detailed information on each compound is provided in the Supplementary Materials.

3.4. Evaluation of anti-inflammatory activity of isolated compounds from mulberry branch bark

The inhibitory effects of the compounds on the inflammatory factors NO, IL-6, and IL-1 β in LPS-induced RAW 264.7 cells were assessed using

Griess and ELISA kits. The experimental results (Figs. 4–6) indicated that all eight compounds inhibited the release of the inflammatory factors NO, IL-6, and IL-1 β to varying extents, with the inhibitory effects increasing in a concentration-dependent manner. Compounds SZP-1, SZP-2, SZP-5, SZP-6, and SZP-8 exhibited the strongest *in vitro* anti-inflammatory activity, with their inhibitory effects on NO, IL-6, and IL-1 β comparable to those of meloxicam at high concentrations.

3.5. Quality control of mulberry branches bark

Previous studies have shown that the chemical constituents of mulberry branch bark are similar to those of mulberry root bark, and the anti-inflammatory activity of mulberry branch bark is comparable to that of mulberry root bark. Therefore, mulberry branch bark could substitute mulberry root bark for anti-inflammatory purposes.

It is well known that the intrinsic quality of traditional Chinese medicines can vary significantly due to geographical conditions and harvesting periods. Therefore, an efficient, rapid, sensitive, and reproducible detection method is essential for ensuring the quality of each batch of herbal materials. The quality of traditional Chinese medicines directly affects clinical efficacy. We conducted a quality control study of

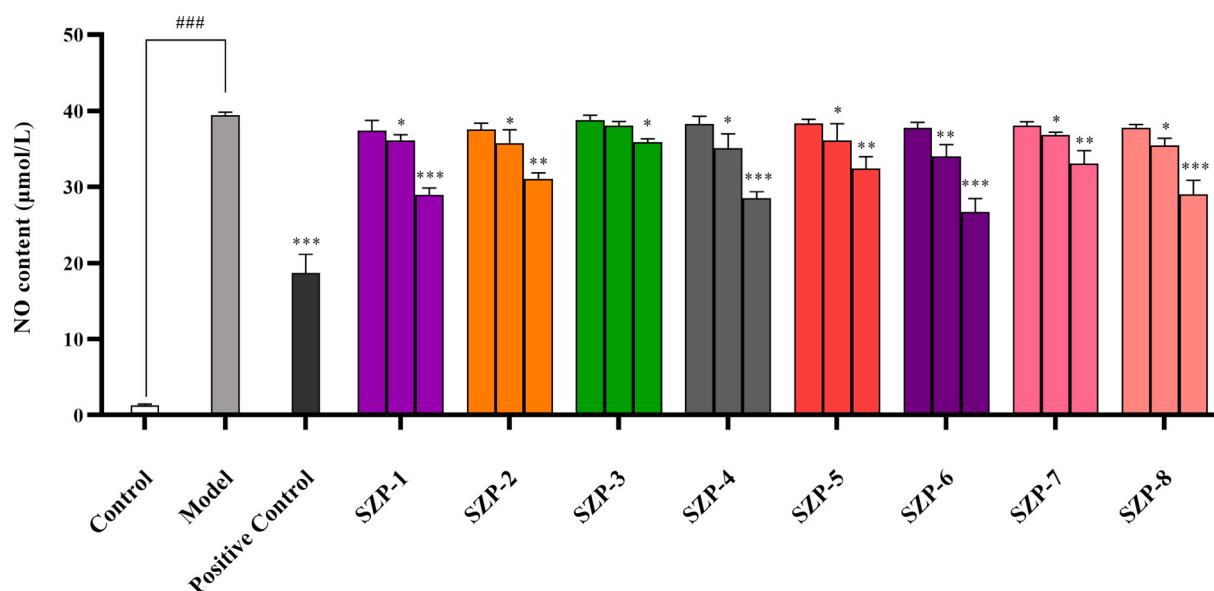


Fig. 4. Inhibitory effects of compounds SZP-1 to SZP-8 on NO content in LPS-induced RAW 264.7 cells. ### $p < 0.001$ vs. the control group; * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$ vs. the model group.

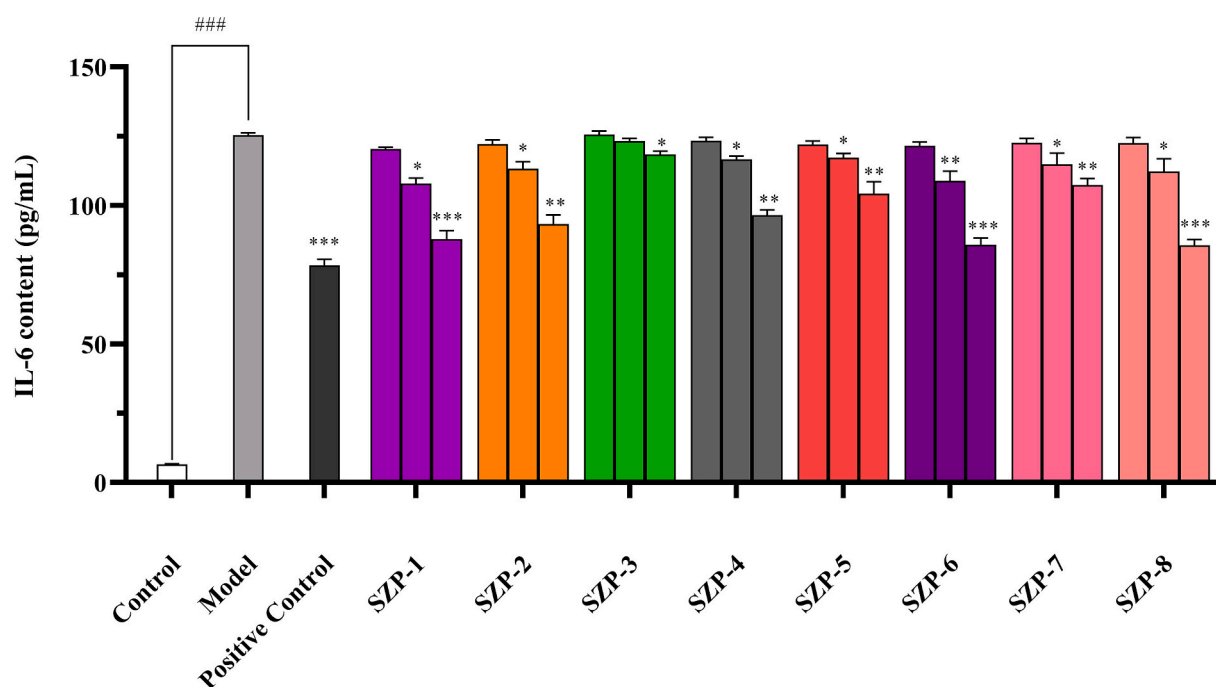


Fig. 5. Inhibitory effects of compounds SZP-1 to SZP-8 on IL-6 content in LPS-induced RAW 264.7 cells. ### $p < 0.001$ vs. the control group; * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$ vs. the model group.

mulberry branch bark to provide reference standards for its medicinal value. Eleven batches of fresh mulberry branches were collected from Jiangxi, Zhejiang, Guangxi, Jiangsu, Guangdong, and Henan provinces. Each batch was identified as the *Morus alba* L. from the Moraceae family. The outer bark of each batch of mulberry branches was peeled off and dried at 45 °C to obtain the mulberry branch bark medicinal material. Following the Chinese Pharmacopoeia (2020 edition), the qualitative identification of mulberry branch was conducted using thin-layer chromatography (TLC) and morphological characteristics. The determination of moisture, ash, and extractive content was primarily performed to ensure the purity and safety of the mulberry branch [3]. The source and batch numbers of mulberry branches are shown in Table 3.

The TLC results for mulberry branch bark are presented in Fig. 7. All 11 batches of medicinal materials displayed multiple bands, indicating similar chemical compositions and consistent spot positions compared to the standard. In the 11 batches of mulberry branch bark, the moisture content ranged from 6.82 % to 8.67 % (average 7.68 %), the total ash content ranged from 3.26 % to 6.08 % (average 4.63 %), and the extractives content ranged from 7.87 % to 9.03 % (average 8.68 %). Additionally, the oxyresveratrol content ranged from 2.775 to 3.154 mg/g, with an average value of 2.946 mg/g, as shown in Table 4. Details of the oxyresveratrol content in mulberry branch bark, along with the accuracy, precision, reproducibility, and stability of its quantification method, are provided in the Supplementary Material.

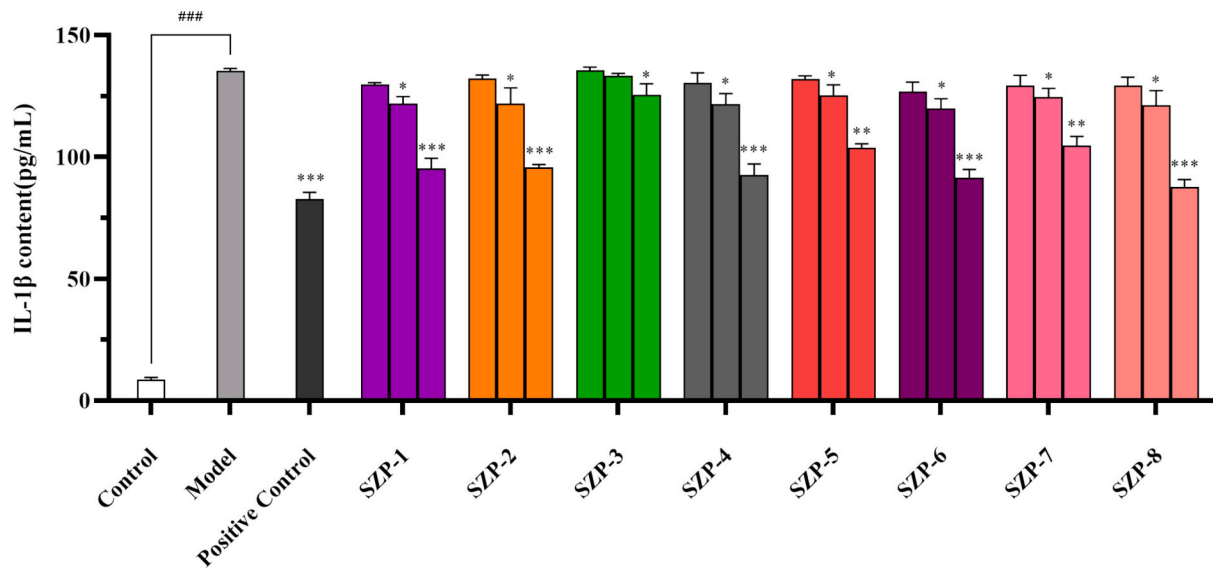


Fig. 6. Inhibitory effects of compounds SZP-1 to SZP-8 on IL-1 β content in LPS-induced RAW 264.7 cells. ### $p < 0.001$ vs. the control group; * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$ vs. the model group.

Table 3
Origins and Batch Numbers of Different Mulberry Branches.

Sample origins	Sample batch number	Sample name
Danyang	20,210,411	Danyang 20,210,411
Guangong	20,210,228	Guangong 20,210,228
Guangxi	20,210,326	Guangxi 20,210,326
Guangxi	20,210,427	Guangxi 20,210,427
Henan	20,210,315	Henan 20,210,315
Huzhou	20,210,310	Huzhou 20,210,310
Huzhou	20,210,325	Huzhou 20,210,325
Jiangxi	20,210,228	Jiangxi 20,210,228
Shandong	20,210,311	Shandong 20,210,311
Zhenjiang	20,210,314	Zhenjiang 20,210,314
Zhenjiang	20,210,411	Zhenjiang 20,210,411

We developed a TLC method using morusin as a control to identify mulberry branch bark, which demonstrated excellent specificity and separation. Based on the assay results, we established quality control standards for mulberry branch bark, including a moisture content not exceeding 10 %, a total ash content of no more than 6 %, and a leachate

content of at least 3 %. Additionally, we developed an HPLC method to quantify oxyresveratrol in mulberry branch bark, which exhibited high specificity and excellent separation. We recommend maintaining the oxyresveratrol content in mulberry branch bark at no less than 0.15 %.

Funding

This work was supported financially by the Guangdong Natural Science Foundation of China (No. 2018A030313907) and the Guangdong Medical Science and Technology Research Fund Project (No. A2023243).

CRediT authorship contribution statement

Wenyuan Li: Writing – review & editing, Visualization, Data curation. **Lin Chen:** Writing – original draft, Methodology, Formal analysis. **Minpeng Li:** Software, Resources, Investigation. **Kaiye Peng:** Software, Formal analysis, Data curation. **Xuemei Lin:** Supervision, Project administration. **Yifan Feng:** Project administration, Funding acquisition. **Yun Zou:** Software, Resources, Investigation. **Xia Wu:** Resources,

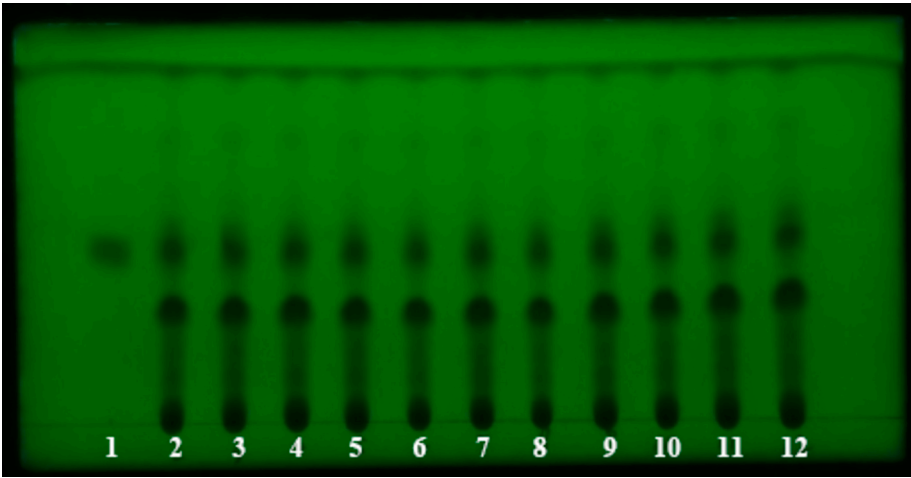


Fig. 7. TLC of 11 Batches of Mulberry Branch Bark and Standards(1: Morusin; 2: Danyang 20,210,411; 3: Guangdong 20,210,228; 4: Guangxi 20,210,326; 5: Guangxi 20,210,427; 6: Henan 20,210,315; 7: Huzhou 20,210,310; 8: Huzhou 20,210,325; 9: Jiangxi 20,210,228; 10: Shandong 20,210,311; 11: Zhenjiang 20,210,314; 12: Zhenjiang 20,210,411).

Table 4
Moisture, Ash, Extractives and Oxyresveratrol Content Analysis of Mulberry Branch Bark (n = 3).

Sample Name	Moisture (%)	Ash (%)	Extractives (%)	Oxyresveratrol (mg/g)
Danyang 20,210,411	8.48	4.67	8.93	2.985
Guangdong 20,210,228	6.93	6.08	8.79	2.893
Guangxi 20,210,326	7.84	5.14	9.03	2.974
Guangxi 20,210,427	7.97	4.77	8.65	2.775
Henan 20,210,315	7.21	5.11	7.87	3.068
Huzhou 20,210,310	6.82	4.36	8.69	2.89
Huzhou 20,210,325	8.67	4.53	8.75	2.868
Jiangxi 20,210,228	6.86	5.04	8.67	2.802
Shandong 20,210,311	8.77	4.56	8.86	2.898
Zhejiang 20,210,314	7.15	3.42	8.62	3.154
Zhejiang 20,210,411	7.82	3.26	8.78	3.104
Average	7.68	4.63	8.68	2.946

Project administration, Methodology, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.fitote.2025.106383>.

Data availability

Data will be made available on request.

References

[1] C.-H. Liu, F. Liu, L. Xiong, Medicinal parts of mulberry (leaf, twig, root bark, and fruit) and compounds thereof are excellent traditional Chinese medicines and foods for diabetes mellitus, *J. Funct. Foods* 106 (2023) 105619, <https://doi.org/10.1016/j.jff.2023.105619>.

[2] R. Dadhwal, R. Banerjee, Ethnopharmacology, pharmacotherapeutics, biomedical and toxicological profile of *Morus alba* L.: a comprehensive review, *S. Afr. J. Bot.* 158 (2023) 98–117, <https://doi.org/10.1016/j.sajb.2023.05.006>.

[3] Chinese Pharmacopoeia Commission, *Pharmacopoeia of the People's Republic of China vol. 1*, China Medical Science Publisher, Beijing, China, 2020.

[4] P. Wen, T.-G. Hu, R.J. Linhardt, S.-T. Liao, H. Wu, Y.-X. Zou, Mulberry: a review of bioactive compounds and advanced processing technology, *Trends Food Sci. Technol.* 83 (2019) 138–158, <https://doi.org/10.1016/j.tifs.2018.11.017>.

[5] J. Hosek, M. Bartos, S. Chudik, S. Dall'Acqua, G. Innocenti, M. Kartal, L. Kokoska, P. Kollar, Z. Kutil, P. Landa, R. Marek, V. Zavalova, M. Zemlicka, K. Smejkal, Natural compound cudraflavone B shows promising anti-inflammatory properties in vitro, *J. Nat. Prod.* 74 (4) (2011) 614–619, <https://doi.org/10.1021/jp100638h>.

[6] N.T. Dat, P.T. Binh, T.P. Quynh Le, H.T. Huong, C.V. Minh, Sanggenon C and O inhibit NO production, iNOS expression and NF-kappaB activation in LPS-induced RAW264.7 cells, *Immunopharmacol. Immunotoxicol.* 34 (1) (2012) 84–88, <https://doi.org/10.3109/08923973.2011.580755>.

[7] Y. Jia, W. He, H. Zhang, L. He, Y. Wang, T. Zhang, J. Peng, P. Sun, Y. Qian, Morusin ameliorates IL-1beta-induced chondrocyte inflammation and osteoarthritis via NF-kB signal pathway, *Drug Des. Devel. Ther.* 14 (2020) 1227–1240, <https://doi.org/10.2147/DDDT.S244462>.

[8] G. Yin, C. Pan, H. Liu, C. Dong, X. Chang, W. Zhou, S. Wang, Z. Du, Oxyresveratrol improves cognitive impairments and episodic-like memory through modulating

Neuroinflammation and PI3K-Akt signaling pathway in LPS-induced mice, *Molecules* 29 (6) (2024), <https://doi.org/10.3390/molecules29061272>.

[9] J. Hao, Y. Gao, J. Xue, Y. Yang, J. Yin, T. Wu, M. Zhang, Phytochemicals, pharmacological effects and molecular mechanisms of mulberry, *Foods* 11 (8) (2022) 1170, <https://doi.org/10.3390/foods11081170>.

[10] S. Wang, X.M. Liu, J. Zhang, Y.Q. Zhang, An efficient preparation of mulberroside A from the branch bark of mulberry and its effect on the inhibition of tyrosinase activity, *PLoS One* 9 (10) (2014) e109396, <https://doi.org/10.1371/journal.pone.0109396>.

[11] Z. Guo, J. Lai, Y. Wu, S. Fang, X. Liang, Investigation on antioxidant activity and different metabolites of mulberry (*Morus* spp.) leaves depending on the harvest months by UPLC-Q-TOF-MS with multivariate tools, *Molecules* 28 (4) (2023) 1947, <https://doi.org/10.3390/molecules28041947>.

[12] G. D'Urso, J.J. Mes, P. Montoro, R.D. Hall, R.C.H. de Vos, Identification of bioactive phytochemicals in mulberries, *Metabolites* 10 (1) (2020) 7, <https://doi.org/10.3390/metabo10010007>.

[13] X.-L. Guo, L. Yang, S.-L. Yu, K. Zhang, J.-H. Wang, H.-Y. Wang, Analysis of secondary metabolites in Xinjiang *Morus nigra* leaves using different extraction methods with UPLC-Q/TOF-MS technology, *Open Chem.* 21 (1) (2023) 20220320, <https://doi.org/10.1515/chem-2022-0320>.

[14] Q.Y. Zhou, X. Liao, H.M. Kuang, J.Y. Li, S.H. Zhang, LC-MS Metabolite Profiling and the Hypoglycemic Activity of *Morus alba* L. Extracts, *Molecules* 27 (17) (2022) 5360, <https://doi.org/10.3390/molecules27175360>.

[15] P.M. Ristivojevic, A. Tahir, F. Malfent, D.M. Opsenica, J.M. Rollinger, High-performance thin-layer chromatography/bioautography and liquid chromatography-mass spectrometry hyphenated with chemometrics for the quality assessment of *Morus alba* samples, *J. Chromatogr. A* 1594 (2019) 190–198, <https://doi.org/10.1016/j.chroma.2019.02.006>.

[16] Q. Ma, Y. Li, R. Zhao, Z. Tang, J. Li, C. Chen, X. Liu, Y. Hu, T. Wang, B. Zhao, Therapeutic mechanisms of mulberry leaves in type 2 diabetes based on metabolomics, *Front. Pharmacol.* 13 (2022) 954477, <https://doi.org/10.3389/fphar.2022.954477>.

[17] X. Sheng, X.-Y. Jia, F. Tang, Y. Wang, A.-J. Hou, The total synthesis of (±)-sanggenol F, *Tetrahedron* 73 (25) (2017) 3485–3491, <https://doi.org/10.1016/j.tet.2017.05.022>.

[18] L. Gao, J. Han, X. Lei, Enantioselective Total syntheses of kuwanon X, Kuwanon Y, and Kuwanol A, *Org. Lett.* 18 (3) (2016) 360–363, <https://doi.org/10.1021/acs.orglett.5b03285>.

[19] Y.-J. Choi, J.I. Lee, M. Fan, Y. Tang, E.-J. Yoon, Y.B. Ryu, E.-K. Kim, Metabolomic analysis of *Morus* cultivar root extracts and their ameliorative effect on testosterone-induced prostate enlargement in Sprague-Dawley rats, *Int. J. Mol. Sci.* 21 (4) (2020) 1435, <https://doi.org/10.3390/ijms21041435>.

[20] M.T. Meenu, G. Kaul, M. Shukla, K.V. Radhakrishnan, S. Chopra, Cudraflavone C from *Artocarpus hirsutus* as a promising inhibitor of pathogenic, multidrug-Resistant *S. Aureus*, Persister, and biofilms: a new insight into a rational explanation of traditional wisdom, *J. Nat. Prod.* 84 (10) (2021) 2700–2708, <https://doi.org/10.1021/acs.jnatprod.1c00578>.

[21] M. Zhu, Z.-J. Wang, Y.-J. He, Y. Qin, Y. Zhou, Z.-H. Qi, Z.-S. Zhou, Y.-Y. Zhu, D.-N. Jin, S.-S. Chen, X.-D. Luo, Bioguided isolation, identification and bioactivity evaluation of anti-MRSA constituents from *Morus alba* Linn, *J. Ethnopharmacol.* 281 (2021) 114542, <https://doi.org/10.1016/j.jep.2021.114542>.

[22] K. Hirakura, T. Fukai, Y. Hano, T. Nomura, Kuwanon W, a natural Diels–Alder type adduct from the root bark of *Morus lhou*, *Phytochemistry* 24 (1) (1985) 159–161, [https://doi.org/10.1016/s0031-9422\(00\)80827-4](https://doi.org/10.1016/s0031-9422(00)80827-4).

[23] W. Jing, R. Yan, Y. Wang, A practical strategy for chemical profiling of herbal medicines using ultra-high performance liquid chromatography coupled with hybrid triple quadrupole-linear ion trap mass spectrometry: a case study of Mori cortex, *Anal. Methods* 7 (2) (2015) 443–457, <https://doi.org/10.1039/c4ay02196g>.

[24] J. Wan, S. Cheng, Y. Wang, C. Wen, Y. Wei, D. Wang, Z. Ouyang, Systematic analysis of bioactive components of distinct medicinal organs of mulberry by high-performance liquid chromatography with electrospray ionization mass spectrometry and molecular docking, *Pharmacogn. Mag.* 17 (75) (2021) 428–437, <https://doi.org/10.4103/pm.pm.533.20>.

[25] Z.-P. Zheng, K.-W. Cheng, Q. Zhu, X.-C. Wang, Z.-X. Lin, M. Wang, Tyrosinase inhibitory constituents from the roots of *Morus nigra*: a structure-activity relationship study, *J. Agric. Food Chem.* 58 (9) (2010) 5368–5373, <https://doi.org/10.1021/jf1003607>.

[26] Y.-H. Lu, C.-N. Lin, H.-H. Ko, S.-Z. Yang, L.-T. Tsao, J.-P. Wang, Two novel and anti-inflammatory constituents of *Artocarpus rigida*, *Helv. Chim. Acta* 85 (6) (2002) 1626–1632, <https://doi.org/10.1002/hlca.200390207>.

[27] T. Fukai, P. Yue-Hu, T. Nomura, X. Chun-Quan, W. Li-Jun, C. Ying-Jie, Components of the root bark of *Morus cathayana*. 1. Structures of five new isoprenylated flavonoids, sanggenols A-E and a diprenyl-2-arylbenzofuran, mulberrofuran V, *Heterocycles* 2 (43) (1996) 425–436, <https://doi.org/10.3987/com-95-7310>.

[28] X. Ren, Y. Sun, Q. Guo, H. Liu, H. Jiang, X. He, X. Li, X. Shi, Z. Xiu, Y. Dong, Ameliorating effect of the Total flavonoids of *Morus nigra* L. on Prediabetic mice based on regulation of inflammation and insulin sensitization, *J. Agric. Food Chem.* 70 (39) (2022) 12484–12501, <https://doi.org/10.1021/acs.jafc.2c04970>.

[29] D. Tungmunthum, A. Thongboonyou, A. Pholboon, A. Yangsabai, Flavonoids and Other Phenolic Compounds from Medicinal Plants for Pharmaceutical and Medical Aspects: An Overview, *Medicines (Basel)* 5 (3) (2018) 93, <https://doi.org/10.3390/medicines5030093>.

[30] X. Xu, Y. Huang, J. Xu, X. He, Y. Wang, Anti-neuroinflammatory and antioxidant phenols from mulberry fruit (*Morus alba* L.), *J. Funct. Foods* 68 (2020) 103914, <https://doi.org/10.1016/j.jff.2020.103914>.